

FUNDACIÓN FULGOR

Undergrad CMOS Course

Homework N°6

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1 Introduction

In this report, the analog *CMOS* circuits designed analytically in *Homework 5* are verified through simulation. The square-law parameters of the previous assignment no longer apply: every transistor is now described by the *BSIM4* models of the open-source *SkyWater sky130A* process design kit (*PDK*) [1]. As required by the assignment, the devices `nfet_01v8` and `pfet_01v8` are used for all transistors.

The simulation environment is the *IIC-OSIC-TOOLS* Docker image [2], which bundles the schematic editor *Xschem* and the circuit simulator *ngspice*. Since recent versions of the image load the *IHP SG13G2 PDK* by default, the active *PDK* must first be switched to *sky130A* with the command `sak-pdk sky130A` [3], launching *Xschem* afterwards from the same terminal. The device models are included in the netlist through the `corner.sym` symbol of the `sky130_fd_pr` library, which adds the corresponding `.lib` statement for the selected process corner [4]. All simulations were carried out in the typical corner (`tt`) at $T = 27\,[°C]$, with the analyses described in the *ngspice* manual [5].

1.1 Transistor Sizing Constraints

The maximum allowed width per finger is $W = 10\,[\mu m]$. Therefore, the widths specified in the exercises are interpreted as required equivalent widths W_{eq}, which are obtained by means of the number of fingers n_f and/or the multiplier m:

$$W_{eq} = W \cdot m \cdot n_f \tag{1}$$

It should be noted that, in the *sky130 Xschem* symbols, the parameter `W` represents the total width of one device, so each finger has a width of W/n_f (this can be seen in the expressions of the `ad`, `as`, `pd` and `ps` parameters of the symbol).

Different combinations of W, m and n_f may produce different results even when they share the same W_{eq}. The main reasons are:

- **Junction parasitics:** Adjacent fingers share their drain and source diffusions, so the diffusion areas and perimeters (`ad`, `as`, `pd`, `ps`) and the associated junction capacitances depend on n_f. With multipliers m, each device is drawn separately and nothing is shared.
- **Width-dependent effects:** The threshold voltage and the mobility change with the width of each individual finger (narrow-width effects), and the layout-dependent stress effects depend on the diffusion lengths around each gate (`sa`, `sb`, `sd`).
- **Binned models:** The *sky130* models are divided into bins of width and length, so different finger widths may be evaluated with different model parameter sets.

- **Gate resistance:** The distributed gate resistance decreases when a wide device is split into several shorter fingers.

These differences are usually small in the *DC* operating point, but they become more noticeable in the small-signal and high-frequency behavior.

2 Exercise 1: Common-Source Bias Design

2.1 Problem Statement

The objective of this exercise is to design the common-source circuit shown in Fig. 1 for a drain current of $I_D = 1\,[mA]$.

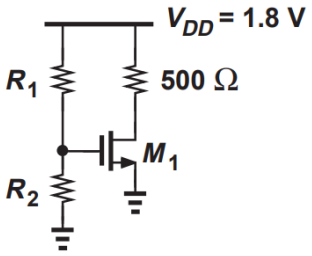


Figure 1: Exercise 1 (Problem 7.2). [6, p. 340]

Given the transistor aspect ratio $W/L = 20/0.18$, we must calculate the bias resistors R_1 and R_2 such that the input impedance is at least $20\,[k\Omega]$, and measure this impedance using an AC analysis at the gate of M_1 at low frequencies. The supply voltage is $V_{DD} = 1.8\,[V]$ and the drain resistor is $R_D = 500\,[\Omega]$.

2.2 Analysis and Resolution

The gate voltage for $I_D = 1\,[mA]$ is first obtained through a DC sweep, then the bias resistors are computed, and finally the complete circuit is verified.

2.2.1 Transistor Sizing

The required equivalent width $W_{eq} = 20\,[\mu m]$ exceeds the maximum width per finger, so according to Eq. 1 the transistor is split into two fingers:

$$\begin{cases} W_{finger} = 10\,[\mu m] \\ n_f = 2 \\ m = 1 \\ L = 0.18\,[\mu m] \end{cases} \implies W_{eq} = 10 \cdot 1 \cdot 2 = 20\,[\mu m] \tag{2}$$

In the `nfet_01v8` symbol this corresponds to `W=20`, `L=0.18`, `nf=2` and `mult=1` (dimensions in $[\mu m]$).

2.2.2 Gate Voltage for $I_D = 1\,[mA]$

In the testbench of Fig. 2, the bias divider is replaced by an ideal source V_{GS} swept from 0 to $1.8\,[V]$, while R_D is kept so that M_1 sees the same V_{DS} as in the final circuit.

Since the gate current is zero, $I_D = -i(V_{DD})$.

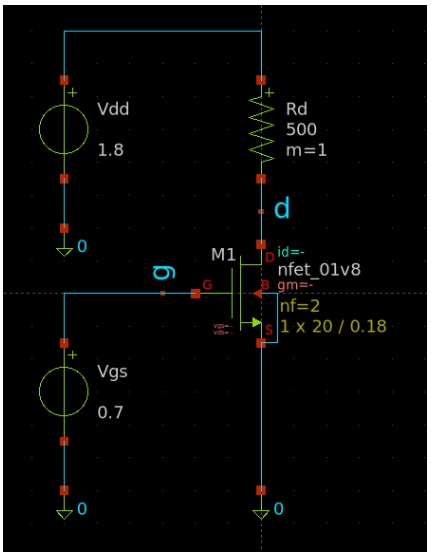


Figure 2: Testbench used to obtain the gate voltage (DC sweep of V_{GS}).

In the control block of Fig. 3, the `meas` statement returns the value of V_G at which $I_D = 1\,[mA]$.

```
.control
save all
dc Vgs 0 1.8 1m
let id = -i(Vdd)
plot id
meas dc VG_1mA find v(g) when id=1m
.endc
```

Figure 3: *ngspice* control block of the DC sweep.

The resulting characteristic is shown in Fig. 4. For high gate voltages the current flattens out at about 3.2 $[mA]$, since M_1 enters the triode region and the current becomes limited by R_D.

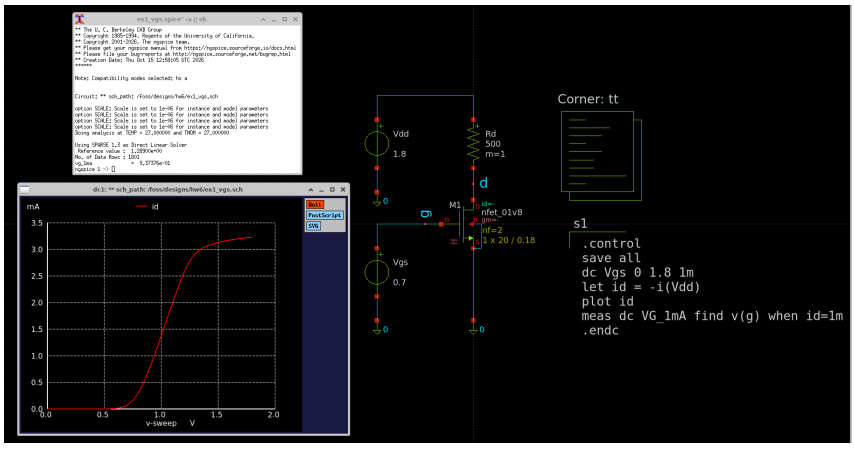


Figure 4: Simulated drain current as a function of V_{GS}.

The measurement yields:

$$V_G = V_{GS}\Big|_{I_D=1\,[mA]} = 0.9374\,[V] \tag{3}$$

2.2.3 Bias Resistors R_1 and R_2

The gate voltage is set by the voltage divider formed by R_1 and R_2, and, at low frequencies, the input impedance is the parallel combination of both resistors (the gate draws no current):

$$\begin{cases} V_G = V_{DD}\dfrac{R_2}{R_1 + R_2} \\ R_{in} = R_1 // R_2 = \dfrac{R_1 R_2}{R_1 + R_2} \end{cases} \tag{4}$$

Solving the system for each resistor:

$$R_1 = R_{in}\frac{V_{DD}}{V_G} \quad , \quad R_2 = R_{in}\frac{V_{DD}}{V_{DD} - V_G} \tag{5}$$

Substituting $R_{in} = 20\,[k\Omega]$ and the simulated gate voltage of Eq. 3:

$$R_1 = 20\,[k\Omega] \cdot \frac{1.8\,[V]}{0.9374\,[V]} \approx 38.41\,[k\Omega] \tag{6}$$

$$R_2 = 20\,[k\Omega] \cdot \frac{1.8\,[V]}{0.8626\,[V]} \approx 41.73\,[k\Omega] \tag{7}$$

2.2.4 Operating Point Verification

The complete circuit of Fig. 5 was then simulated. The current source I_0 is only used for the input impedance measurement and, since its DC value is zero, it does not modify the bias point.

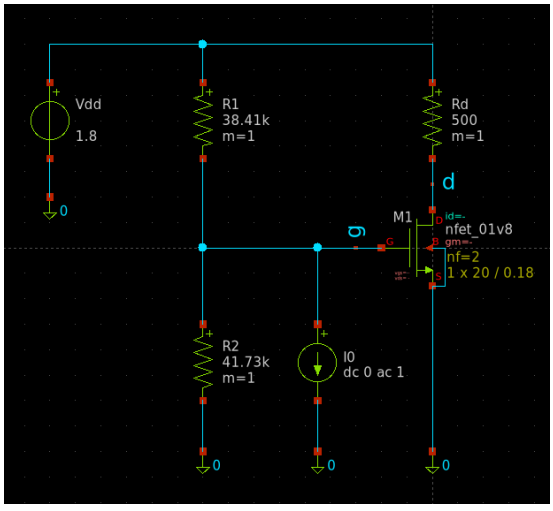


Figure 5: Final circuit with the bias resistors and the AC test current source.

The control block of Fig. 6 performs an operating point analysis (`op`) and prints the node voltages together with the internal parameters of M_1. Since the *sky130* transistors are implemented as subcircuits, the instance `M1` of the schematic appears in *ngspice* as `m.xm1.msky130_fd_pr__nfet_01v8`.

```
.control
save all
save @m.xm1.msky130_fd_pr__nfet_01v8[id] @m.xm1.msky130_fd_pr__nfet_01v8[gm]
save @m.xm1.msky130_fd_pr__nfet_01v8[vth] @m.xm1.msky130_fd_pr__nfet_01v8[vdsat]
op
print v(g) v(d)
print @m.xm1.msky130_fd_pr__nfet_01v8[id] @m.xm1.msky130_fd_pr__nfet_01v8[gm]
print @m.xm1.msky130_fd_pr__nfet_01v8[vth] @m.xm1.msky130_fd_pr__nfet_01v8[vdsat]
ac dec 20 1 10G
let zin = mag(v(g))
plot zin xlog ylog
meas ac Zin_1k find zin at=1k
.endc
```

Figure 6: *ngspice* control block of the operating point and AC analyses.

The simulated operating point is summarized in Tab. 1.

Parameter	**Simulated value**
V_G	$0.9373\,[V]$
V_D	$1.3003\,[V]$
I_D	$0.9995\,[mA]$
g_m	$6.794\,[mA/V]$
V_{Th}	$0.6981\,[V]$
$V_{D,sat}$	$0.1663\,[V]$

Table 1: Simulated operating point of M_1.

The drain current matches the target within $0.1\,\%$. In short-channel devices, the saturation voltage $V_{D,sat}$ given by the model is smaller than $V_{OV} = V_{GS} - V_{Th} \approx 0.239\,[V]$ due to velocity saturation, so saturation is verified against $V_{D,sat}$:

$$V_{DS} = 1.3003\,[V] > V_{D,sat} = 0.1663\,[V] \tag{8}$$

Thus, M_1 operates in saturation.

2.2.5 Input Impedance Measurement

The current source I_0 (`dc 0 ac 1`) injects a unit AC current into the gate, so the magnitude of the gate voltage equals the input impedance:

$$|Z_{in}(f)| = \frac{|V_g(f)|}{|I_0|} = \frac{|V_g(f)|}{1\,[A]} \tag{9}$$

The result of an AC sweep from $1\,[Hz]$ to $10\,[GHz]$ is shown in Fig. 7.

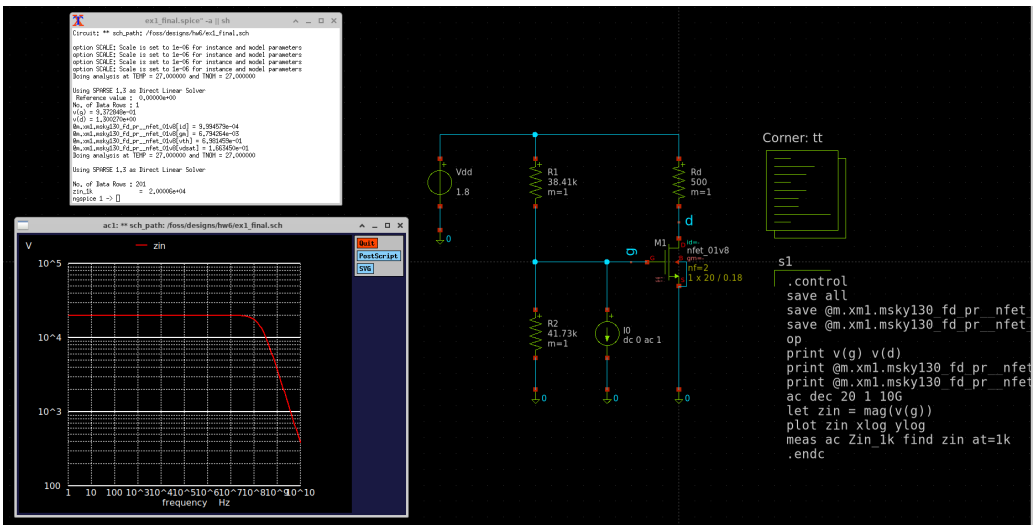


Figure 7: Simulated input impedance magnitude as a function of frequency.

At low frequencies the input impedance is constant and equal to $R_1//R_2$. The measured value at $1\,[kHz]$ is:

$$|Z_{in}|_{f=1\,[kHz]} = 20.0006\,[k\Omega] \geq 20\,[k\Omega] \tag{10}$$

Thus, the requirement is fulfilled. Above tens of megahertz the impedance decreases, since the gate capacitances of M_1 become comparable to the bias resistors; this is why it must be measured at low frequencies.

3 Exercise 2: Common-Source with Source Degeneration

3.1 Problem Statement

The circuit shown in Fig. 8 must be designed for a voltage drop of $V_{RS} = 200\,[mV]$ across R_S.

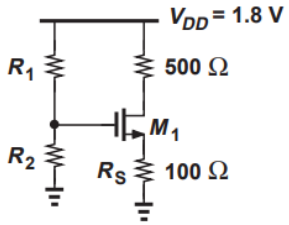


Figure 8: Exercise 2 (Problem 7.4). [6, p. 341]

The specific requirements are:

(a) Calculate the minimum allowable value of W/L if M_1 must remain in saturation.

(b) Calculate the required values of R_1 and R_2 if the input impedance must be at least $30\,[k\Omega]$, measuring it with an AC analysis at the gate of M_1 at low frequencies.

The circuit parameters are $V_{DD} = 1.8\,[V]$, $R_D = 500\,[\Omega]$ and $R_S = 100\,[\Omega]$, as shown in Fig. 8.

3.2 Analysis and Resolution

The bias point is partially fixed by the circuit, regardless of the transistor model:

$$\begin{cases} I_D = \dfrac{V_{RS}}{R_S} = \dfrac{0.2\,[V]}{100\,[\Omega]} = 2\,[mA] \\ V_S = V_{RS} = 0.2\,[V] \\ V_D = V_{DD} - I_D R_D = 1.8 - (2 \times 10^{-3})(500) = 0.8\,[V] \\ V_{DS} = V_D - V_S = 0.6\,[V] \end{cases} \tag{11}$$

It should be noted that the bulk of the `nfet_01v8` device is the p-type substrate, so it is connected to ground and $V_{SB} = 0.2\,[V]$, which increases the threshold voltage due to the body effect.

3.2.1 Part (a): Minimum allowable value of W/L

Here the unknown is the transistor width. The testbench of Fig. 9 forces M_1 into the required operating point with ideal sources ($V_D = 0.8\,[V]$, $V_S = 0.2\,[V]$) while the gate voltage is swept, using a reference width $W_{ref} = 20\,[\mu m]$ ($n_f = 2$). Since, for the same terminal voltages, the drain current is approximately proportional to the width, the minimum width is obtained afterwards by scaling.

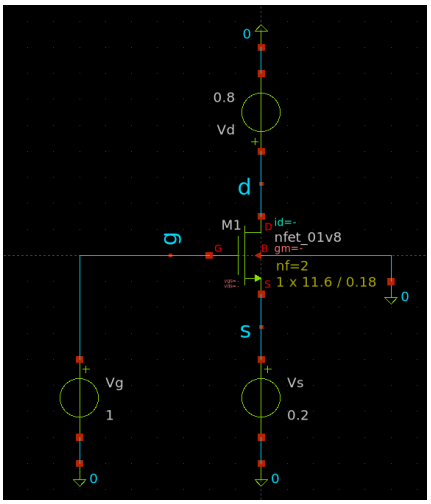


Figure 9: Testbench used to determine the minimum width (terminals forced to the required operating point).

Two saturation criteria were evaluated: the model condition $V_{DS} \geq V_{D,sat}$, and the classical condition $V_{DS} \geq V_{OV}$, with the overdrive voltage computed from the threshold voltage given by the model:

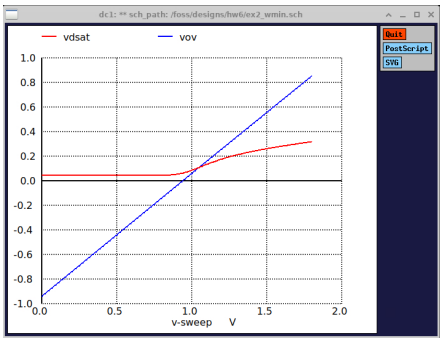
$$V_{OV} = V_{GS} - V_{Th} = V_G - V_S - V_{Th} \tag{12}$$

The control block used for this simulation is shown in Fig. 10.

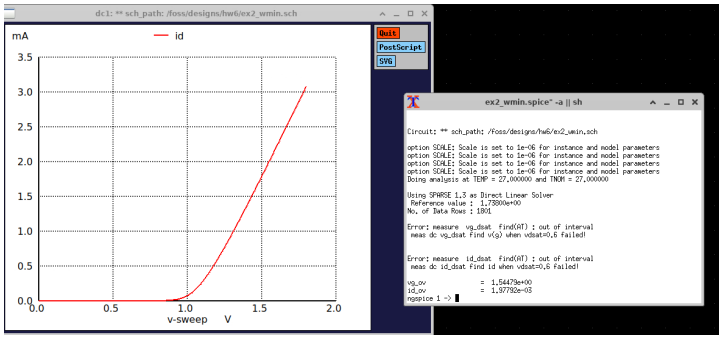
```
.control
save all
save @m.xm1.msky130_fd_pr__nfet_01v8[vdsat] @m.xm1.msky130_fd_pr__nfet_01v8[vth]
dc Vg 0 1.8 1m
let id = -i(Vd)
let vdsat = @m.xm1.msky130_fd_pr__nfet_01v8[vdsat]
let vov = v(g) - v(s) - @m.xm1.msky130_fd_pr__nfet_01v8[vth]
plot vdsat vov
plot id
meas dc VG_dsat find v(g) when vdsat=0.6
meas dc ID_dsat find id when vdsat=0.6
meas dc VG_ov find v(g) when vov=0.6
meas dc ID_ov find id when vov=0.6
.endc
```

Figure 10: *ngspice* control block for the minimum width analysis.

The results for the reference width are shown in Fig. 11.



(a) $V_{D,sat}$ and V_{OV} versus V_G.



(b) I_D versus V_G.

Figure 11: Simulation results for the reference width $W_{ref} = 20\,[\mu m]$, with $V_S = 0.2\,[V]$ and $V_D = 0.8\,[V]$.

Fig. 11a shows that, due to velocity saturation, $V_{D,sat}$ levels off at about $0.32\,[V]$ and never reaches $0.6\,[V]$, so the model criterion imposes no restriction on the width. Therefore, the classical criterion $V_{DS} \geq V_{OV}$ was adopted, which is the condition stated by the problem and is conservative with respect to the actual $V_{D,sat}$.

The overdrive voltage reaches $0.6\,[V]$ at the following point:

$$V_G = 1.5484\,[V] \qquad , \qquad I_{D,ref} = 3.450\,[mA] \tag{13}$$

Scaling the reference width to the required current:

$$W_{min} \approx W_{ref} \frac{I_D}{I_{D,ref}} = 20\,[\mu m] \cdot \frac{2\,[mA]}{3.450\,[mA]} \approx 11.59\,[\mu m] \tag{14}$$

To validate the linear scaling, the simulation was repeated with $W = 11.6\,[\mu m]$ ($n_f = 2$), obtaining $V_G = 1.5448\,[V]$ and $I_D = 1.978\,[mA]$ at the boundary $V_{OV} = 0.6\,[V]$. The error of approximately $1.1\,\%$ is caused by the reduction of the width per finger from 10 to $5.8\,[\mu m]$ (see subsection 1.1). Correcting the width with this result:

$$W_{min} = 11.6\,[\mu m] \cdot \frac{2\,[mA]}{1.978\,[mA]} \approx 11.73\,[\mu m] \implies \left(\frac{W}{L}\right)_{min} = \frac{11.73}{0.18} \approx 65.2 \tag{15}$$

To avoid operating exactly at the edge of the saturation region, the width was rounded up to $W = 12\,[\mu m]$, implemented with $n_f = 2$ fingers of $6\,[\mu m]$:

$$\left(\frac{W}{L}\right) = \frac{12}{0.18} \approx 66.7 \qquad (W = 12,\ n_f = 2,\ m = 1) \tag{16}$$

3.2.2 Part (b): Required values of R_1 and R_2

With the selected width, the gate voltage for $I_D = 2\,[mA]$ was obtained with the testbench of Fig. 12, using a control block analogous to the one of Fig. 3.

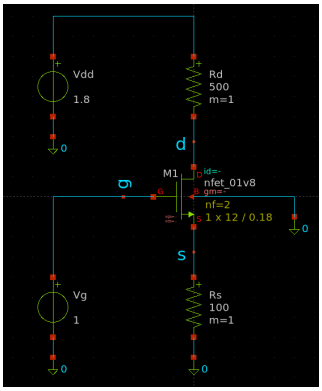


Figure 12: Testbench used to obtain the gate voltage for $I_D = 2\,[mA]$.

The measurements yield:

$$V_G = 1.5343\,[V] \qquad , \qquad V_S = 0.200\,[V] \tag{17}$$

The source voltage confirms the required drop across R_S, and V_G is slightly lower than in Eq. 13 because $W > W_{min}$.

Applying Eq. 5 with $R_{in} = 30\,[k\Omega]$:

$$R_1 = 30\,[k\Omega] \cdot \frac{1.8\,[V]}{1.5343\,[V]} \approx 35.20\,[k\Omega] \tag{18}$$

$$R_2 = 30\,[k\Omega] \cdot \frac{1.8\,[V]}{0.2657\,[V]} \approx 203.2\,[k\Omega] \tag{19}$$

3.2.3 Operating Point Verification

The complete circuit of Fig. 13 was simulated with the same control block of Fig. 6, also printing $v(s)$ and V_{OV}.

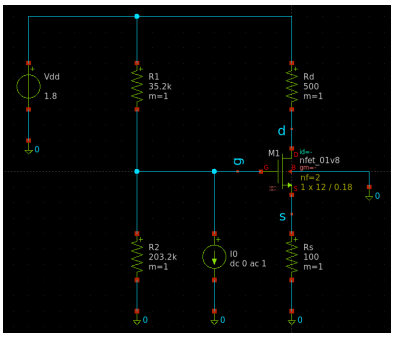


Figure 13: Final circuit with the bias resistors and the AC test current source.

The simulated operating point is summarized in Tab. 2.

Parameter	Simulated value
V_G	$1.5342\,[V]$
V_S	$0.2000\,[V]$
V_D	$0.8000\,[V]$
I_D	$1.9999\,[mA]$
g_m	$4.782\,[mA/V]$
V_{Th}	$0.7454\,[V]$
V_{OV}	$0.5888\,[V]$
$V_{D,sat}$	$0.2688\,[V]$

Table 2: Simulated operating point of M_1.

The drain current and the voltage drop across R_S match the design targets. The saturation condition is satisfied with both criteria:

$$V_{DS} = 0.6000\,[V] > V_{OV} = 0.5888\,[V] > V_{D,sat} = 0.2688\,[V] \tag{20}$$

The margin with respect to the classical criterion is small, as expected, and the threshold voltage is $47\,[mV]$ higher than in section 2 due to the body effect.

3.2.4 Input Impedance Measurement

Following subsubsection 2.2.5, the input impedance shown in Fig. 14 was obtained.

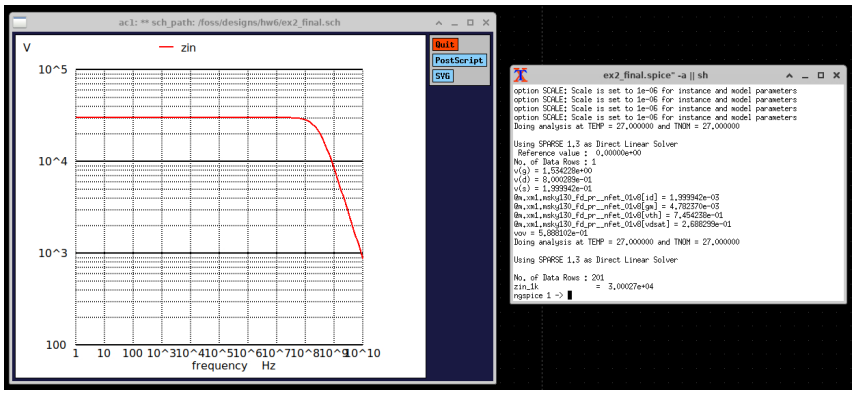


Figure 14: Simulated input impedance magnitude as a function of frequency.

The measured value at $1\,[kHz]$ is:

$$|Z_{in}|_{f=1\,[kHz]} = 30.003\,[k\Omega] \geq 30\,[k\Omega] \tag{21}$$

Thus, the input impedance requirement is fulfilled.

4 Exercise 3: Self-Biased Stage Design

4.1 Problem Statement

The self-biased stage shown in Fig. 15 must be designed for a drain current of $I_D = 1\,[mA]$.

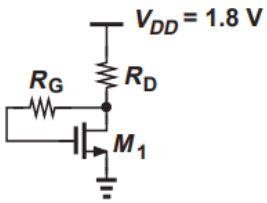


Figure 15: Exercise 3 (Problem 7.6). [6, p. 341]

If transistor M_1 must provide a transconductance of $g_m = 1/(100\,[\Omega])$, we need to calculate the required value of R_D, neglecting R_G. The supply voltage is $V_{DD} = 1.8\,[V]$.

4.2 Analysis and Resolution

Since the *DC* gate current is zero, there is no voltage drop across R_G, so $V_G = V_D$ and M_1 behaves as a diode-connected transistor. As $V_{DS} = V_{GS}$ and $V_{GS} > V_{GS} - V_{Th}$, the transistor always operates in saturation.

The problem does not specify the aspect ratio of M_1, and with the *PDK* models there is no closed-form expression, so its width must be designed. Keeping $L = 0.18\,[\mu m]$, the specification can be expressed in terms of the transconductance efficiency:

$$\frac{g_m}{I_D} = \frac{10\,[mA/V]}{1\,[mA]} = 10\,[V^{-1}] \tag{22}$$

In the g_m/I_D methodology, for a given length, g_m/I_D depends only on the current density I_D/W, so once the density that yields $10\,[V^{-1}]$ is known, the width follows from the required current.

4.2.1 Transconductance Efficiency Characterization

The testbench of Fig. 16 was used to characterize a diode-connected reference transistor with $W_{ref} = 100\,[\mu m]$, implemented with $n_f = 10$ fingers of $10\,[\mu m]$. As in the final circuit, gate and drain are tied together and driven by a source V_{GS} swept from 0.3 to $1.8\,[V]$.

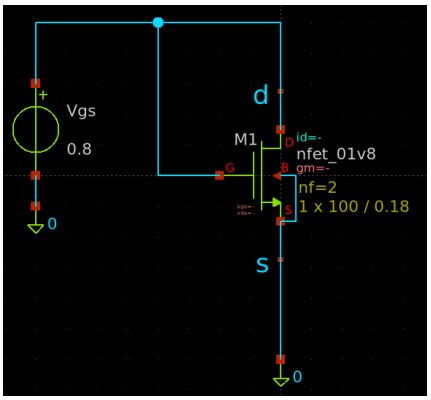


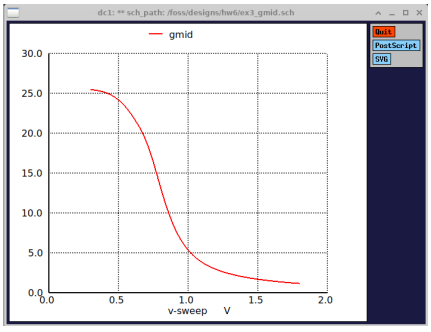
Figure 16: Testbench for the g_m/I_D characterization of the diode-connected reference transistor.

The control block of Fig. 17 computes the transconductance efficiency from the g_m reported by the model and measures the operating point where it equals $10\,[V^{-1}]$.

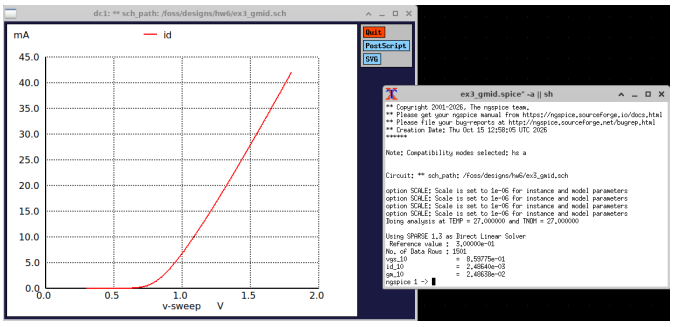
```
.control
save all
save @m.xm1.msky130_fd_pr__nfet_01v8[gm]
dc Vgs 0.3 1.8 1m
let id = -i(Vgs)
let gm = @m.xm1.msky130_fd_pr__nfet_01v8[gm]
let gmid = gm/id
plot gmid
plot id
meas dc VGS_10 find v(d) when gmid=10
meas dc ID_10 find id when gmid=10
meas dc GM_10 find gm when gmid=10
.endc
```

Figure 17: *ngspice* control block for the g_m/I_D characterization.

The results are shown in Fig. 18.



(a) g_m/I_D versus V_{GS}.



(b) I_D versus V_{GS}.

Figure 18: Characterization of the diode-connected reference transistor ($W_{ref} = 100\,[\mu m]$, $L = 0.18\,[\mu m]$).

Fig. 18a shows the typical behavior of the transconductance efficiency: in weak inversion it reaches about $25.5\,[V^{-1}]$, close to the theoretical limit $1/(nU_T) \approx 26\,[V^{-1}]$ (with

$n \approx 1.5$ and $U_T \approx 25.9\,[mV]$), and it decreases monotonically as the transistor moves into strong inversion. The required value of $10\,[V^{-1}]$ lies in the moderate inversion region. The measurements at this point yield:

$$V_{GS} = 0.8598\,[V] \quad , \quad I_{D,ref} = 2.486\,[mA] \quad , \quad g_{m,ref} = 24.86\,[mA/V] \tag{23}$$

Scaling the reference width to the required current:

$$W = W_{ref}\frac{I_D}{I_{D,ref}} = 100\,[\mu m] \cdot \frac{1\,[mA]}{2.486\,[mA]} \approx 40.2\,[\mu m] \implies \frac{W}{L} \approx 223 \tag{24}$$

The width was set to $W = 40\,[\mu m]$ ($n_f = 4$ fingers of $10\,[\mu m]$), keeping the same width per finger as the reference transistor.

4.2.2 Width Verification and R_D Calculation

To verify the selected width, the diode-connected transistor was biased with an ideal current source of exactly $1\,[mA]$, as shown in Fig. 19. An operating point analysis was performed, printing the same device parameters as in Fig. 6.

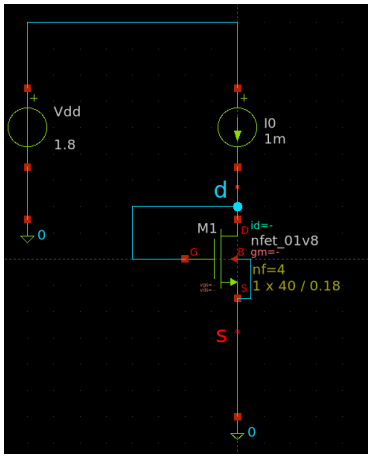


Figure 19: Diode-connected transistor ($W = 40\,[\mu m]$, $n_f = 4$) biased with an ideal $1\,[mA]$ current source.

The simulation yields $V_{GS} = 0.8613\,[V]$ and $g_m = 9.956\,[mA/V]$, that is, an error of $0.44\,\%$ with respect to the specification, caused by rounding the width down from 40.2 to $40\,[\mu m]$. Given this gate-to-source voltage, the drain resistor follows from *Ohm's law*, since $V_D = V_{GS}$:

$$R_D = \frac{V_{DD} - V_{GS}}{I_D} = \frac{1.8\,[V] - 0.8613\,[V]}{1 \times 10^{-3}\,[A]} \approx 938.7\,[\Omega] \tag{25}$$

4.2.3 Final Circuit Verification

The complete circuit of Fig. 20 was simulated with $R_D = 938.7\,[\Omega]$ and $R_G = 100\,[k\Omega]$. Since no DC current flows through R_G, its value does not affect the bias point.

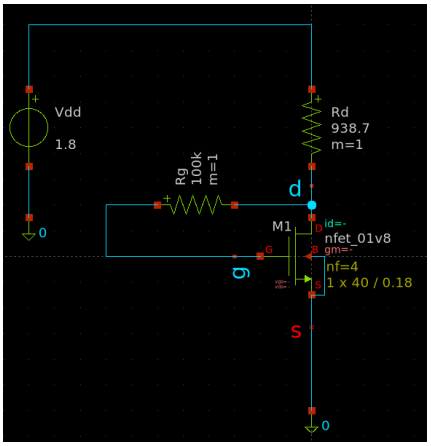


Figure 20: Final self-biased circuit.

The simulated operating point is summarized in Tab. 3.

Parameter	Simulated value
V_D	$0.8613\,[V]$
V_G	$0.8613\,[V]$
I_D	$1.0000\,[mA]$
g_m	$9.956\,[mA/V]$
g_m/I_D	$9.956\,[V^{-1}]$
V_{Th}	$0.6999\,[V]$
$V_{D,sat}$	$0.1322\,[V]$

Table 3: Simulated operating point of M_1.

The gate and drain voltages are identical, confirming that R_G carries no DC current, and the saturation condition is amply satisfied:

$$V_{DS} = 0.8613\,[V] > V_{D,sat} = 0.1322\,[V] \tag{26}$$

5 Exercise 4: Common-Source Stage Analysis

5.1 Problem Statement

In the common-source stage shown in Fig. 21, the transistor has an aspect ratio of $W/L = 30/0.18$.

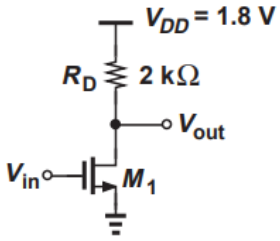


Figure 21: Exercise 4 (Problem 7.17). [6, p. 342]

We are tasked with determining:

(a) The gate voltage V_G that yields a drain current of $I_D = 0.5\,[mA]$, and verifying that M_1 operates in saturation.

(b) The low-frequency voltage gain A_v of the stage under this bias condition.

The parameters given are $V_{DD} = 1.8\,[V]$ and $R_D = 2\,[k\Omega]$.

5.2 Analysis and Resolution

The required equivalent width $W_{eq} = 30\,[\mu m]$ was implemented with $n_f = 3$ fingers of $10\,[\mu m]$ ($W = 30$, $L = 0.18$, $n_f = 3$). As in the previous exercises, the drain voltage is fixed by the circuit and does not depend on the transistor model:

$$V_D = V_{DD} - I_D R_D = 1.8 - (0.5 \times 10^{-3})(2 \times 10^3) = 0.8\,[V] \tag{27}$$

5.2.1 Part (a): Gate Voltage and Saturation Verification

The gate voltage was obtained with the testbench of Fig. 22, in which the input source V_{in} is swept from 0 to $1.8\,[V]$ with R_D connected. The control block is analogous to the one of Fig. 3, measuring the input and output voltages at the point where $I_D = 0.5\,[mA]$.

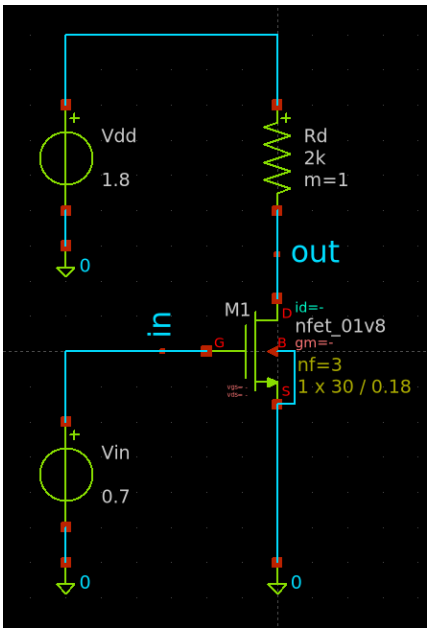


Figure 22: Testbench used to obtain the gate voltage for $I_D = 0.5\,[mA]$.

The resulting characteristic is shown in Fig. 23; for high gate voltages the transistor enters the triode region and the current is limited to about $V_{DD}/R_D = 0.9\,[mA]$.

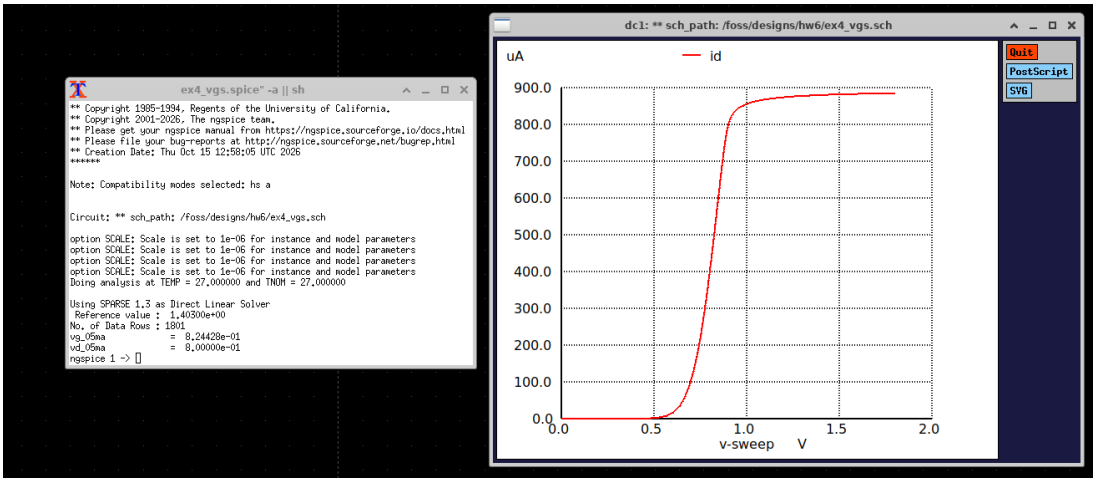


Figure 23: Simulated drain current as a function of V_{in}.

The measurements at $I_D = 0.5\,[mA]$ yield:

$$V_G = 0.8244\,[V] \qquad , \qquad V_D = 0.800\,[V] \tag{28}$$

Setting the input source to this value in the final testbench of subsubsection 5.2.2, the operating point of Tab. 4 was obtained.

Parameter	Simulated value
V_G	$0.8244\,[V]$
V_D	$0.8000\,[V]$
I_D	$0.5000\,[mA]$
g_m	$5.968\,[mA/V]$
g_{ds}	$160.1\,[\mu A/V]$
$r_o = 1/g_{ds}$	$6.244\,[k\Omega]$
V_{Th}	$0.6999\,[V]$
V_{OV}	$0.1245\,[V]$
$V_{D,sat}$	$0.1143\,[V]$

Table 4: Simulated operating point of M_1.

The saturation condition is amply satisfied with both criteria:

$$V_{DS} = 0.800\,[V] > V_{OV} = 0.1245\,[V] > V_{D,sat} = 0.1143\,[V] \tag{29}$$

5.2.2 Part (b): Voltage Gain

Since the *PDK* models include channel-length modulation, the output resistance $r_o = 1/g_{ds}$ of M_1 appears in parallel with R_D. The small-signal voltage gain of the stage is then:

$$A_v = -g_m\,(R_D // r_o) = -\frac{g_m}{g_{ds} + 1/R_D} \tag{30}$$

The gain was obtained both from an *AC* analysis and from Eq. 30 with the model parameters, using the testbench of Fig. 24. Its input source (`dc 0.824428 ac 1`) sets the bias point and applies a unit *AC* signal, so $|V_{out}| = |A_v|$.

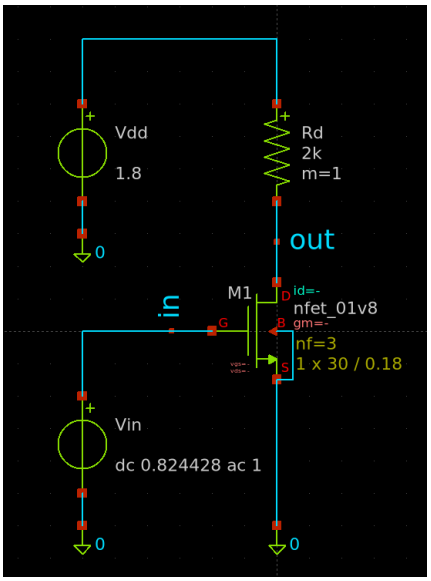


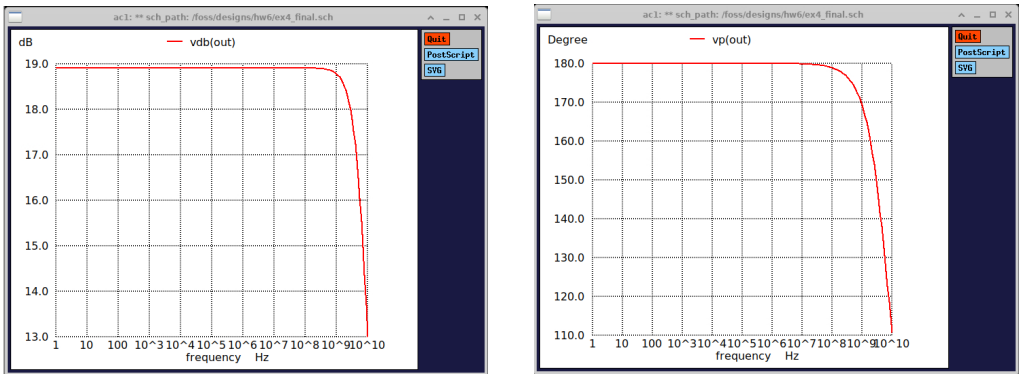
Figure 24: Final common-source stage with the *DC* + *AC* input source.

The control block used is shown in Fig. 25.

```
.control
save all
save @m.xm1.msky130_fd_pr__nfet_01v8[id] @m.xm1.msky130_fd_pr__nfet_01v8[gm]
save @m.xm1.msky130_fd_pr__nfet_01v8[gds] @m.xm1.msky130_fd_pr__nfet_01v8[vth]
save @m.xm1.msky130_fd_pr__nfet_01v8[vdsat]
op
print v(in) v(out)
print @m.xm1.msky130_fd_pr__nfet_01v8[id] @m.xm1.msky130_fd_pr__nfet_01v8[gm]
print @m.xm1.msky130_fd_pr__nfet_01v8[gds] @m.xm1.msky130_fd_pr__nfet_01v8[vth]
print @m.xm1.msky130_fd_pr__nfet_01v8[vdsat]
let av_calc = @m.xm1.msky130_fd_pr__nfet_01v8[gm]/(@m.xm1.msky130_fd_pr__nfet_01v8[gds] + 1/2000)
print av_calc
ac dec 20 1 10G
let av = mag(v(out))
set units=degrees
plot vdb(out) xlog
plot vp(out) xlog
meas ac Av_1k find av at=1k
.endc
```

Figure 25: *ngspice* control block of the operating point and AC analyses.

The frequency response of the stage is shown in Fig. 26.



(a) Magnitude $[dB]$. (b) Phase $[°]$.

Figure 26: Simulated frequency response V_{out}/V_{in} of the common-source stage.

At low frequencies the magnitude is constant and the phase is $180\,[°]$, as expected for an inverting stage. The gain measured at $1\,[kHz]$ is:

$$A_{v,AC} = -8.821 \left[\frac{V}{V}\right] \qquad (18.91\,[dB]) \tag{31}$$

On the other hand, substituting the values of Tab. 4 into Eq. 30:

$$A_{v,calc} = -\frac{5.968 \times 10^{-3}}{160.1 \times 10^{-6} + 500 \times 10^{-6}} = -\frac{5.968 \times 10^{-3}}{660.1 \times 10^{-6}} \approx -9.041 \left[\frac{V}{V}\right] \tag{32}$$

Both results differ by about $2.4\,\%$, since Eq. 30 only uses the intrinsic g_m and g_{ds}, whereas the AC analysis linearizes the complete device model, including parasitic elements such as the source and drain series resistances.

6 Exercise 5: Common-Source Stage with Active Load

6.1 Problem Statement

The common-source stage shown in Fig. 27 must provide a low-frequency voltage gain of $|A_v| = 10$ with a bias current of $I_D = 0.5\,[mA]$.

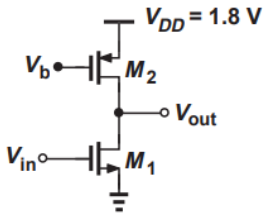


Figure 27: Exercise 5 (Problem 7.20). [6, p. 343]

Since the tool does not provide the parameter r_o, it is computed as $r_o = 1/g_{ds}$, so the gain of the stage is:

$$|A_v| = g_{m1}\,(r_{o1}//r_{o2}) = \frac{g_{m1}}{g_{ds1} + g_{ds2}} \tag{33}$$

We are asked to:

(a) Calculate the required value of $(W/L)_1$.

(b) Calculate the required value of V_B if $(W/L)_2 = 20/0.18$.

6.2 Analysis and Resolution

The output conductances given by the models depend on the drain-to-source voltage of each transistor, so the output bias voltage was set first to $V_{out} = V_{DD}/2 = 0.9\,[V]$, which gives a symmetrical saturation margin to both transistors.

Since Eq. 33 requires g_{ds2}, the load M_2 (whose size is given) is analyzed first. The bulk of the *PMOS* device (n-well) is connected to V_{DD}.

6.2.1 Part (b): Load Transistor and Bias Voltage V_B

The load M_2 ($W = 20\,[\mu m]$, $n_f = 2$) was simulated alone with the testbench of Fig. 28, with its drain forced to $0.9\,[V]$ while V_B is swept.

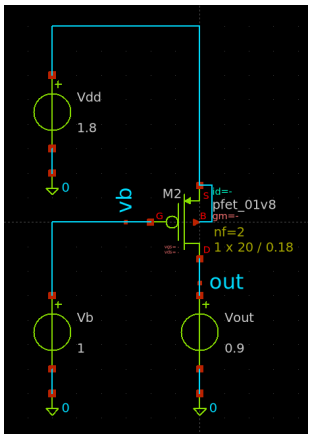


Figure 28: Testbench of the *PMOS* load with its drain forced to $V_{out} = 0.9\,[V]$.

The control block of Fig. 29 measures V_B, g_{ds2} and $V_{D,sat2}$ at $I_{D2} = 0.5\,[mA]$.

```
.control
save all
save @m.xm2.msky130_fd_pr__pfet_01v8[gds] @m.xm2.msky130_fd_pr__pfet_01v8[vdsat]
dc Vb 0 1.8 1m
let id2 = i(Vout)
let gds2 = @m.xm2.msky130_fd_pr__pfet_01v8[gds]
let vdsat2 = @m.xm2.msky130_fd_pr__pfet_01v8[vdsat]
plot id2
meas dc VB_05 find v(vb) when id2=0.5m
meas dc GDS2_05 find gds2 when id2=0.5m
meas dc VDSAT2_05 find vdsat2 when id2=0.5m
.endc
```

Figure 29: *ngspice* control block for the *PMOS* load characterization.

The resulting current is shown in Fig. 30.

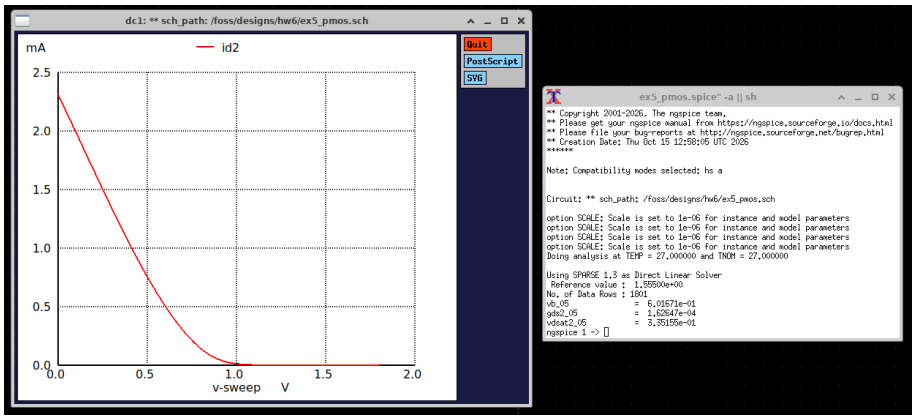


Figure 30: Drain current of M_2 as a function of V_B, with $V_{out} = 0.9\,[V]$.

The measurements at $I_{D2} = 0.5\,[mA]$ yield:

$$V_B = 0.6017\,[V] \quad , \quad g_{ds2} = 162.6\,[\mu A/V] \quad , \quad V_{D,sat2} = 0.3352\,[V] \tag{34}$$

Therefore, the required bias voltage is $V_B = 0.6017\,[V]$ ($V_{SG2} = 1.198\,[V]$), with $r_{o2} \approx 6.148\,[k\Omega]$ and M_2 in saturation.

6.2.2 Part (a): Required $(W/L)_1$

Since both transistors carry the same current $I_D = 0.5\,[mA]$, the gain condition of Eq. 33 can be divided by I_D and rewritten in terms of quantities normalized to the drain current:

$$\frac{g_{m1}}{I_D} - 10\,\frac{g_{ds1}}{I_D} = 10\,\frac{g_{ds2}}{I_D} \tag{35}$$

The left-hand side depends only on the current density of M_1, as in the g_m/I_D methodology of section 4, while the right-hand side is a constant obtained from Eq. 34:

$$10\,\frac{g_{ds2}}{I_D} = 10 \cdot \frac{162.6 \times 10^{-6}}{0.5 \times 10^{-3}} \approx 3.253\,[V^{-1}] \tag{36}$$

A reference M_1 ($W_{ref} = 20\,[\mu m]$, $n_f = 2$) with its drain forced to $0.9\,[V]$ (Fig. 31) was swept, computing $f = g_{m1}/I_D - 10\,g_{ds1}/I_D$ with the control block of Fig. 32.

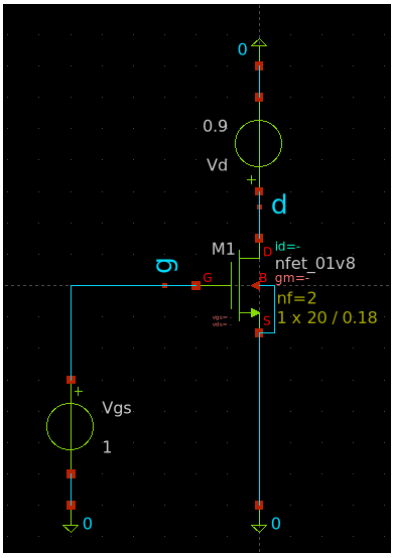


Figure 31: Testbench of the reference *NMOS* transistor with its drain forced to $0.9\,[V]$.

```
.control
save all
save @m.xm1.msky130_fd_pr__nfet_01v8[gm] @m.xm1.msky130_fd_pr__nfet_01v8[gds]
dc Vgs 0.5 1.8 1m
let id1 = -i(Vd)
let gmid = @m.xm1.msky130_fd_pr__nfet_01v8[gm]/id1
let gdsid = @m.xm1.msky130_fd_pr__nfet_01v8[gds]/id1
let f = gmid - 10*gdsid
plot gmid f
meas dc VGS_Q find v(g) when f=3.2529
meas dc ID_Q find id1 when f=3.2529
meas dc GMID_Q find gmid when f=3.2529
.endc
```

Figure 32: *ngspice* control block for the sizing of M_1.

The curves of g_{m1}/I_D and f are shown in Fig. 33.

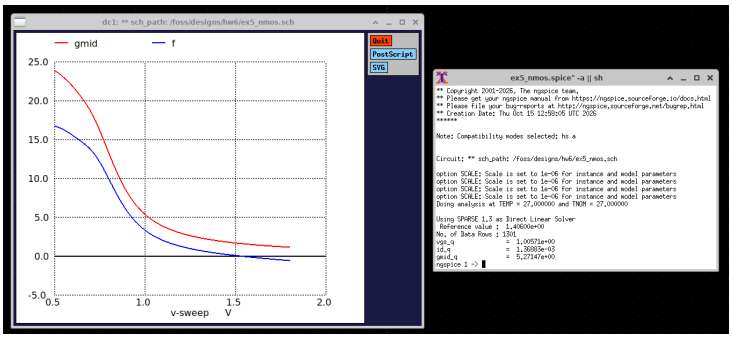


Figure 33: g_m/I_D and $f = g_m/I_D - 10\, g_{ds}/I_D$ of the reference *NMOS* transistor as a function of V_{GS}.

The measurements at the solution point yield:

$$V_{GS} = 1.0057\,[V] \quad , \quad I_{D,ref} = 1.369\,[mA] \quad , \quad \frac{g_{m1}}{I_D} = 5.271\,[V^{-1}] \tag{37}$$

At this point $g_{ds1}/I_D \approx 0.202\,[V^{-1}]$, lower than $g_{ds2}/I_D \approx 0.325\,[V^{-1}]$, so the *PMOS* device has a lower output resistance. Scaling the reference width:

$$W_1 = W_{ref}\frac{I_D}{I_{D,ref}} = 20\,[\mu m] \cdot \frac{0.5\,[mA]}{1.369\,[mA]} \approx 7.3\,[\mu m] \implies \left(\frac{W}{L}\right)_1 = \frac{7.3}{0.18} \approx 40.6 \tag{38}$$

Since W_1 is lower than the maximum width per finger, M_1 was implemented with a single finger ($W = 7.3$, $n_f = 1$).

6.2.3 Final Circuit and Operating Point

The complete stage of Fig. 34 was simulated using the bias voltage obtained in subsubsection 6.2.1, that is, $V_B = 0.601671\,[V]$.

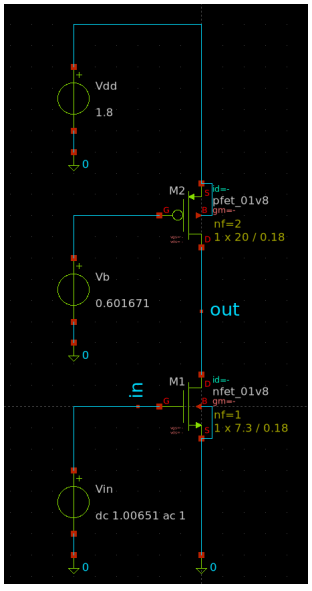


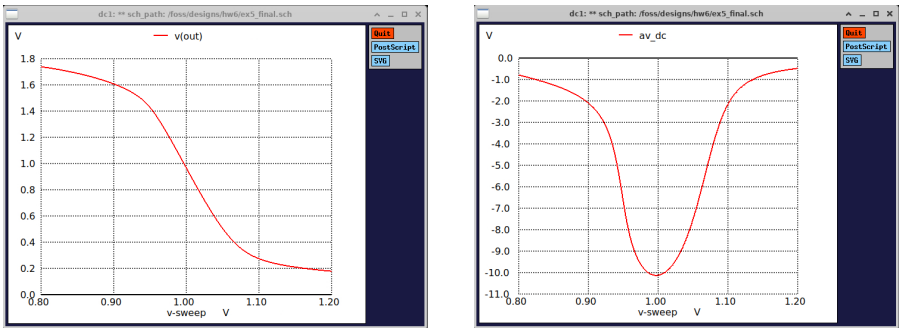
Figure 34: Final common-source stage with active *PMOS* load.

Since the high-impedance output node is very sensitive to the input voltage, the input bias was found with a fine DC sweep of V_{in}, measuring the point where $V_{out} = 0.9\,[V]$ and the slope of the transfer characteristic (Fig. 35).

```
.control
save all
dc Vin 0.8 1.2 0.1m
let av_dc = deriv(v(out))
plot v(out)
plot av_dc
meas dc VIN_Q find v(in) when v(out)=0.9
meas dc AV_DC find av_dc when v(out)=0.9
.endc
```

Figure 35: *ngspice* control block of the DC transfer characteristic.

The transfer characteristic and its slope are shown in Fig. 36; the high-gain region corresponds to both transistors in saturation.



(a) V_{out} versus V_{in}. (b) dV_{out}/dV_{in} versus V_{in}.

Figure 36: Simulated DC transfer characteristic of the stage and its slope.

The measurements at $V_{out} = 0.9\,[V]$ yield:

$$V_{in,Q} = 1.0065\,[V] \quad , \quad A_{v,DC} = \left.\frac{dV_{out}}{dV_{in}}\right|_{V_{out}=0.9\,[V]} = -10.05\,\left[\frac{V}{V}\right] \tag{39}$$

The input bias agrees with Eq. 37 within $1\,[mV]$, which validates the scaling.

Effect of the finger configuration. In a first attempt, M_1 was inadvertently simulated with $W = 7.3\,[\mu m]$ and $n_f = 2$, that is, with two fingers of $3.65\,[\mu m]$. Tab. 5 compares both configurations, which have the same equivalent width.

Configuration	**Width per finger**	$\mathbf{V_{in,Q}}$	$\mathbf{A_{v,DC}}$
$n_f = 2$	$3.65\,[\mu m]$	$1.0151\,[V]$	$-9.54\,[V/V]$
$n_f = 1$	$7.30\,[\mu m]$	$1.0065\,[V]$	$-10.05\,[V/V]$

Table 5: Effect of the finger configuration of M_1 for the same $W_{eq} = 7.3\,[\mu m]$.

With narrower fingers, the transistor needs $9\,[mV]$ more gate voltage and the gain drops by about 5 %, a direct example of the question raised in subsection 1.1.

Finally, with the input source set to `dc 1.00651 ac 1`, the operating point of Tab. 6 and the AC response were obtained with a control block analogous to Fig. 25.

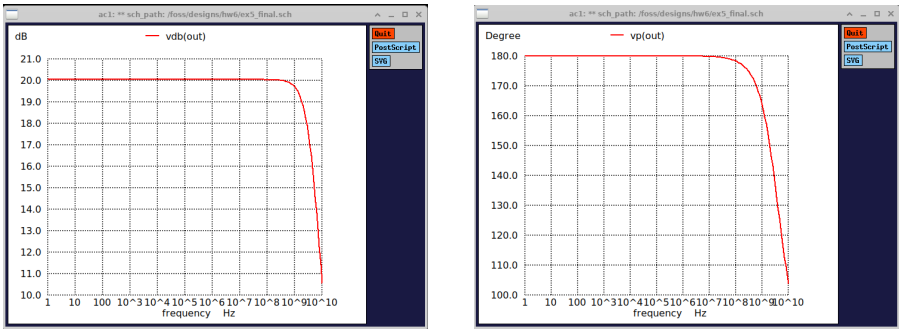
Parameter	Simulated value
I_D	$0.5000\,[mA]$
g_{m1}	$2.631\,[mA/V]$
g_{ds1} (r_{o1})	$100.9\,[\mu A/V]$ $(9.913\,[k\Omega])$
g_{ds2} (r_{o2})	$162.6\,[\mu A/V]$ $(6.148\,[k\Omega])$
$V_{D,sat1}$	$0.1918\,[V]$
$V_{D,sat2}$	$0.3352\,[V]$

Table 6: Simulated operating point of M_1 and M_2.

Both transistors operate in saturation, since $V_{DS1} = 0.9\,[V] > V_{D,sat1} = 0.1918\,[V]$ and $V_{SD2} = 0.9\,[V] > V_{D,sat2} = 0.3352\,[V]$.

6.2.4 Voltage Gain

The frequency response of the stage is shown in Fig. 37.



(a) Magnitude $[dB]$. (b) Phase [°].

Figure 37: Simulated frequency response V_{out}/V_{in} of the stage with active load.

At low frequencies the magnitude is constant at approximately $20\,[dB]$ with a phase of $180\,[°]$. The gain measured at $1\,[kHz]$ is:

$$A_{v,AC} = -10.05\,\left[\frac{V}{V}\right] \qquad (20.04\,[dB]) \tag{40}$$

which coincides with the slope of the transfer characteristic of Eq. 39. On the other hand, substituting the values of Tab. 6 into Eq. 33:

$$|A_{v,calc}| = \frac{2.631 \times 10^{-3}}{100.9 \times 10^{-6} + 162.6 \times 10^{-6}} = \frac{2.631 \times 10^{-3}}{263.5 \times 10^{-6}} \approx 9.984\,\left[\frac{V}{V}\right] \tag{41}$$

The difference is below $1\,\%$, so the gain requirement $|A_v| = 10$ is fulfilled.

7 Exercise 6: Common-Source Stage with Diode-Connected Load

7.1 Problem Statement

The circuit shown in Fig. 38 must be designed for a voltage gain of $|A_v| = 3$. Given $(W/L)_1 = 20/0.18$, we must determine $(W/L)_2$.

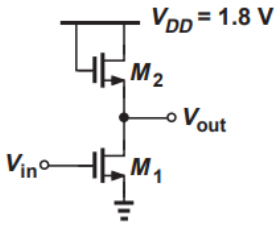


Figure 38: Exercise 6 (Problem 7.25). [6, p. 344]

7.2 Analysis and Resolution

The load M_2 of Fig. 38 is an *NMOS* transistor with its gate and drain at V_{DD} and its source at the output. Neglecting body effect and with $\lambda = 0$, the square-law model gives:

$$|A_v| = \frac{g_{m1}}{g_{m2}} = \sqrt{\frac{(W/L)_1}{(W/L)_2}} \implies \left(\frac{W}{L}\right)_2 = \frac{(W/L)_1}{|A_v|^2} = \frac{20/0.18}{9} \approx 12.35 \tag{42}$$

With the *PDK* models, the bulk of M_2 is grounded while its source is at V_{out}, so the body transconductance g_{mb2} appears, and the output conductances are no longer zero. The gain is then:

$$|A_v| = \frac{g_{m1}}{g_{m2} + g_{mb2} + g_{ds1} + g_{ds2}} \tag{43}$$

Since the bias current is not specified, it was set to $I_D = 0.5\,[mA]$, so the output voltage is fixed by the load, $V_{out} = V_{DD} - V_{GS2}$. As in section 6, M_1 is characterized first and then M_2 is sized.

7.2.1 Input Transistor Characterization

M_1 ($W = 20\,[\mu m]$, $n_f = 2$) was simulated with its drain forced to an estimate of V_{out} (Fig. 39), measuring g_{m1} and g_{ds1} at $I_D = 0.5\,[mA]$ (Fig. 40).

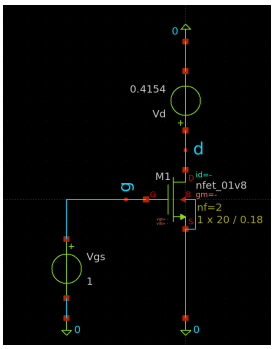


Figure 39: Testbench of the input transistor with its drain forced to the estimated output voltage.

```
.control
save all
save @m.xm1.msky130_fd_pr__nfet_01v8[gm] @m.xm1.msky130_fd_pr__nfet_01v8[gds]
save @m.xm1.msky130_fd_pr__nfet_01v8[vdsat]
dc Vgs 0.3 1.2 0.1m
let id1 = -i(Vd)
let gm1 = @m.xm1.msky130_fd_pr__nfet_01v8[gm]
let gds1 = @m.xm1.msky130_fd_pr__nfet_01v8[gds]
let vdsat1 = @m.xm1.msky130_fd_pr__nfet_01v8[vdsat]
plot id1
meas dc VGS1_Q find v(g) when id1=0.5m
meas dc GM1_Q find gm1 when id1=0.5m
meas dc GDS1_Q find gds1 when id1=0.5m
meas dc VDSAT1_Q find vdsat1 when id1=0.5m
.endc
```

Figure 40: *ngspice* control block for the characterization of M_1.

Once g_{m1} and g_{ds1} are known, Eq. 43 gives the conductance that the load must present, which, divided by the bias current, becomes a target value for the load transistor:

$$\frac{g_{m2} + g_{mb2} + g_{ds2}}{I_D} = \frac{g_{m1}/3 - g_{ds1}}{I_D} \tag{44}$$

Since V_{out} is not known in advance, the design was iterated starting from $V_{out} = 0.5\,[V]$, as summarized in Tab. 7.

Parameter	**Iteration 1**	**Iteration 2**
V_{DS1} (forced)	$0.5000\,[V]$	$0.4154\,[V]$
V_{GS1}	$0.8728\,[V]$	$0.8763\,[V]$
g_{m1}	$4.831\,[mA/V]$	$4.767\,[mA/V]$
g_{ds1}	$186.3\,[\mu A/V]$	$217.2\,[\mu A/V]$
$V_{D,sat1}$	$0.1372\,[V]$	$0.1388\,[V]$
Target of Eq. 44	$2.848\,[V^{-1}]$	$2.743\,[V^{-1}]$
$I_{D,ref}$ $(W_{ref} = 2\,[\mu m])$	$0.3739\,[mA]$	$0.3915\,[mA]$
V_{out}	$0.4154\,[V]$	$0.3976\,[V]$

Table 7: Iterations of the design of the stage with diode-connected load.

The second iteration changed V_{out} by only $18\,[mV]$, so the design was considered converged, with M_1 in saturation in both cases.

7.2.2 Load Transistor Sizing

A diode-connected reference load with $W_{ref} = 2\,[\mu m]$ and grounded bulk was characterized by sweeping the current drawn from its source (Fig. 41).

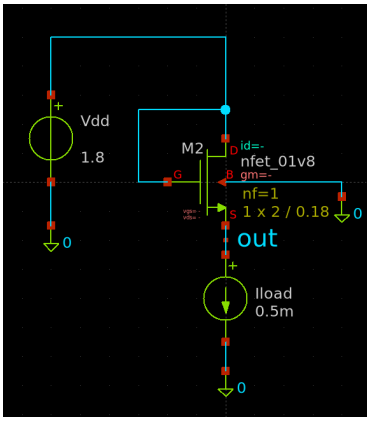


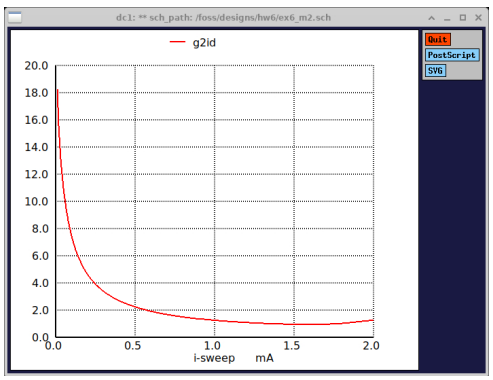
Figure 41: Testbench of the diode-connected load transistor with its bulk grounded.

For each current, the control block of Fig. 42 computes the normalized load conductance and measures the point where it equals the target of Eq. 44.

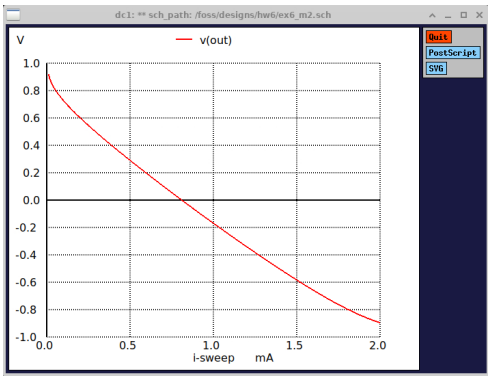
```
.control
save all
save @m.xm2.msky130_fd_pr__nfet_01v8[gm] @m.xm2.msky130_fd_pr__nfet_01v8[gmbs]
save @m.xm2.msky130_fd_pr__nfet_01v8[gds] @m.xm2.msky130_fd_pr__nfet_01v8[vth]
dc Iload 10u 2m 1u
let id2 = -i(Vdd)
let g2 = @m.xm2.msky130_fd_pr__nfet_01v8[gm] + @m.xm2.msky130_fd_pr__nfet_01v8[gmbs] + @m.xm2.msky130_fd_pr__nfet_01v8[gds]
let g2id = g2/id2
plot g2id
plot v(out)
meas dc I_REF find id2 when g2id=2.743
meas dc VOUT_REF find v(out) when g2id=2.743
.endc
```

Figure 42: *ngspice* control block for the sizing of M_2 (second iteration).

The results of the second iteration are shown in Fig. 43; the region with negative V_{out} has no physical meaning and was not used.



(a) $(g_{m2} + g_{mb2} + g_{ds2})/I_D$ versus I_D.



(b) V_{out} versus I_D.

Figure 43: Characterization of the diode-connected reference load ($W_{ref} = 2\,[\mu m]$, $L = 0.18\,[\mu m]$).

At the solution point, $I_{D,ref} = 0.3915\,[mA]$ and $V_{out} = 0.3976\,[V]$. Since scaling at constant current density keeps V_{GS2} and the body effect unchanged:

$$W_2 = W_{ref}\frac{I_D}{I_{D,ref}} = 2\,[\mu m]\cdot\frac{0.5\,[mA]}{0.3915\,[mA]} \approx 2.55\,[\mu m] \implies \left(\frac{W}{L}\right)_2 = \frac{2.55}{0.18} \approx 14.2 \quad (45)$$

The load transistor was implemented with a single finger ($W = 2.55$, $n_f = 1$).

7.2.3 Final Circuit and Operating Point

The complete stage of Fig. 44 was first simulated with a *DC* sweep of V_{in}, using a control block analogous to the one of Fig. 35 but measuring the point where $I_D = 0.5\,[mA]$.

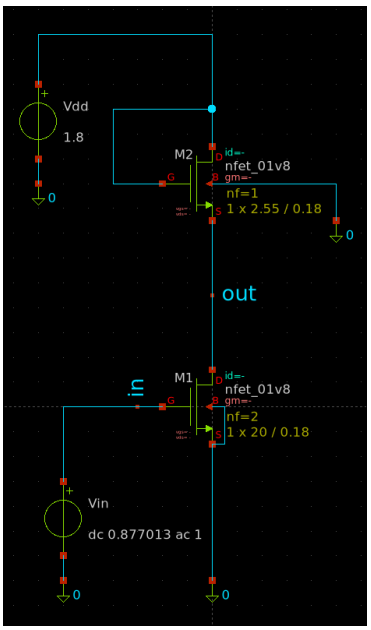
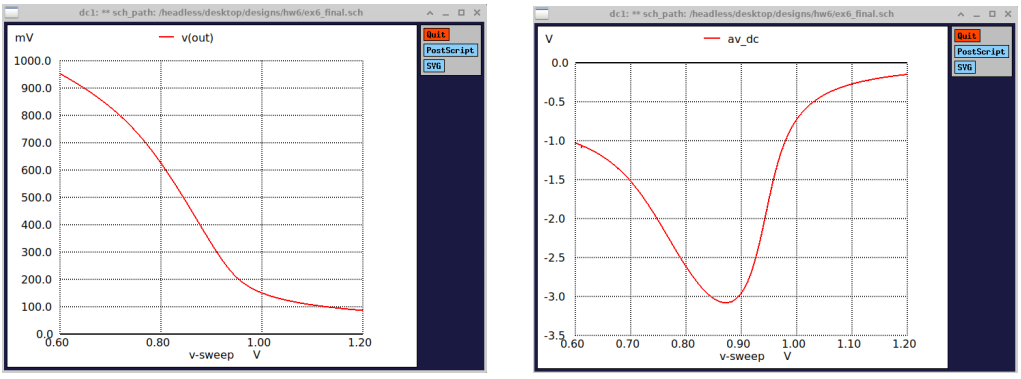


Figure 44: Final common-source stage with diode-connected *NMOS* load.

The transfer characteristic and its slope are shown in Fig. 45.



(a) V_{out} versus V_{in}.

(b) dV_{out}/dV_{in} versus V_{in}.

Figure 45: Simulated *DC* transfer characteristic of the stage and its slope.

The measurements at $I_D = 0.5\,[mA]$ yield:

$$V_{in,Q} = 0.8770\,[V] \quad , \quad V_{out,Q} = 0.4004\,[V] \quad , \quad A_{v,DC} = -3.076\,\left[\frac{V}{V}\right] \tag{46}$$

Unlike the square-law prediction of Eq. 42, Fig. 45b shows that the gain depends strongly on the bias point: it decreases at low currents, where g_{m2}/I_D increases, and collapses at high currents, where M_1 enters the triode region.

With the input source set to `dc 0.877013 ac 1`, the operating point of Tab. 8 was obtained.

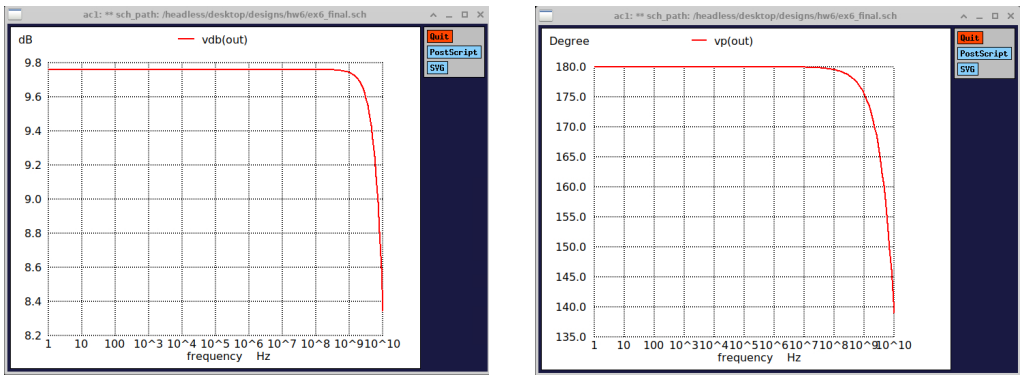
Parameter	Simulated value
I_D	$0.5000\,[mA]$
V_{out}	$0.4004\,[V]$
g_{m1}	$4.752\,[mA/V]$
g_{ds1}	$225.0\,[\mu A/V]$
$V_{D,sat1}$	$0.1391\,[V]$
g_{m2}	$1.131\,[mA/V]$
g_{mb2}	$160.7\,[\mu A/V]$
g_{ds2}	$76.04\,[\mu A/V]$
V_{Th2}	$0.8148\,[V]$

Table 8: Simulated operating point of M_1 and M_2.

M_1 operates in saturation, since $V_{DS1} = 0.4004\,[V] > V_{D,sat1} = 0.1391\,[V]$, while M_2 is always saturated due to its diode connection. The threshold voltage of M_2 is about $115\,[mV]$ higher than that of a device with grounded source (section 2), as a consequence of the body effect with $V_{SB2} \approx 0.4\,[V]$. The ratio $g_{mb2}/g_{m2} \approx 0.14$ quantifies the contribution of the body effect to the load conductance.

7.2.4 Voltage Gain

The frequency response of the stage is shown in Fig. 46.



(a) Magnitude $[dB]$. (b) Phase $[°]$.

Figure 46: Simulated frequency response V_{out}/V_{in} of the stage with diode-connected load.

At low frequencies the magnitude is constant at approximately $9.76\,[dB]$ with a phase of $180\,[°]$. The gain measured at $1\,[kHz]$ is:

$$A_{v,AC} = -3.076\,\left[\frac{V}{V}\right] \qquad (9.76\,[dB]) \tag{47}$$

which coincides with the slope of the transfer characteristic of Eq. 46. On the other hand, substituting the values of Tab. 8 into Eq. 43:

$$|A_{v,calc}| = \frac{4.752 \times 10^{-3}}{(1.131 + 0.1607 + 0.2250 + 0.0760) \times 10^{-3}} = \frac{4.752}{1.593} \approx 2.983\,\left[\frac{V}{V}\right] \tag{48}$$

Both results differ by about $3\,\%$, so the gain requirement is fulfilled. Considering only the transconductances, the gain would be $g_{m1}/g_{m2} \approx 4.20$; the output conductances reduce it to 3.32, and the body effect to the value of Eq. 48.

8 Conclusion

In this report, the six circuits of *Homework 5* were redesigned and verified with the *sky130A* models in *Xschem* and *ngspice*. All designs met their specifications: the bias currents and input impedances were obtained within 0.1 %, and the gains of the active-load and diode-connected-load stages reached 10.05 and 3.08, respectively. Tab. 9 summarizes the main design values compared with the hand calculations of *Homework 5*.

Exercise	Parameter	Homework 5	Homework 6
1	V_G R_1 / R_2	$0.700\,[V]$ 51.43 / $32.73\,[k\Omega]$	$0.937\,[V]$ 38.41 / $41.73\,[k\Omega]$
2	$(W/L)_{min}$ R_1 / R_2	55.56 45 / $90\,[k\Omega]$	65.2 35.20 / $203.2\,[k\Omega]$
3	R_D	$1.2\,[k\Omega]$	$938.7\,[\Omega]$
4	V_G $\lvert A_v \rvert$	$0.573\,[V]$ 11.54	$0.824\,[V]$ 8.82
5	$(W/L)_1$ V_B	7.81 $1.100\,[V]$	40.6 $0.602\,[V]$
6	$(W/L)_2$	24.69	14.2

Table 9: Main design values of *Homework 5* (square-law model) and *Homework 6* (*sky130A*).

The differences are explained by a few physical effects that the square-law model neglects. The threshold voltage of the real devices is about $0.7\,[V]$ instead of $0.4\,[V]$, which raises every gate voltage and changes the bias dividers. Velocity saturation and mobility degradation make $V_{D,sat}$ much smaller than V_{OV} and the relation $g_m = 2I_D/V_{OV}$ only approximate, especially in moderate inversion. The output resistance at $L = 0.18\,[\mu m]$ is finite (equivalent λ between 0.2 and $0.33\,[V^{-1}]$, about twice the values assumed), which reduced the gain of Exercise 4 and required a much larger M_1 in Exercise 5. The body effect, ignored in *Homework 5*, raised the threshold voltage in Exercises 2 and 6 and reduced the gain of the diode-connected load. In addition, in Exercise 6 the load is an *NMOS* device, whereas *Homework 5* treated it as *PMOS*; the corrected square-law value (12.35) is close to the simulated one.

Finally, the g_m/I_D methodology combined with width scaling at constant current density proved to be an efficient way to size transistors directly from the models, and the experiment of Exercise 5 showed that the finger configuration must also be taken into account. Hand calculations remain useful as a first estimate, but the final design values must be obtained from the *PDK* models.

9 Bibliography

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